# Oral and Maxillofacial Radiology
# MRI & US

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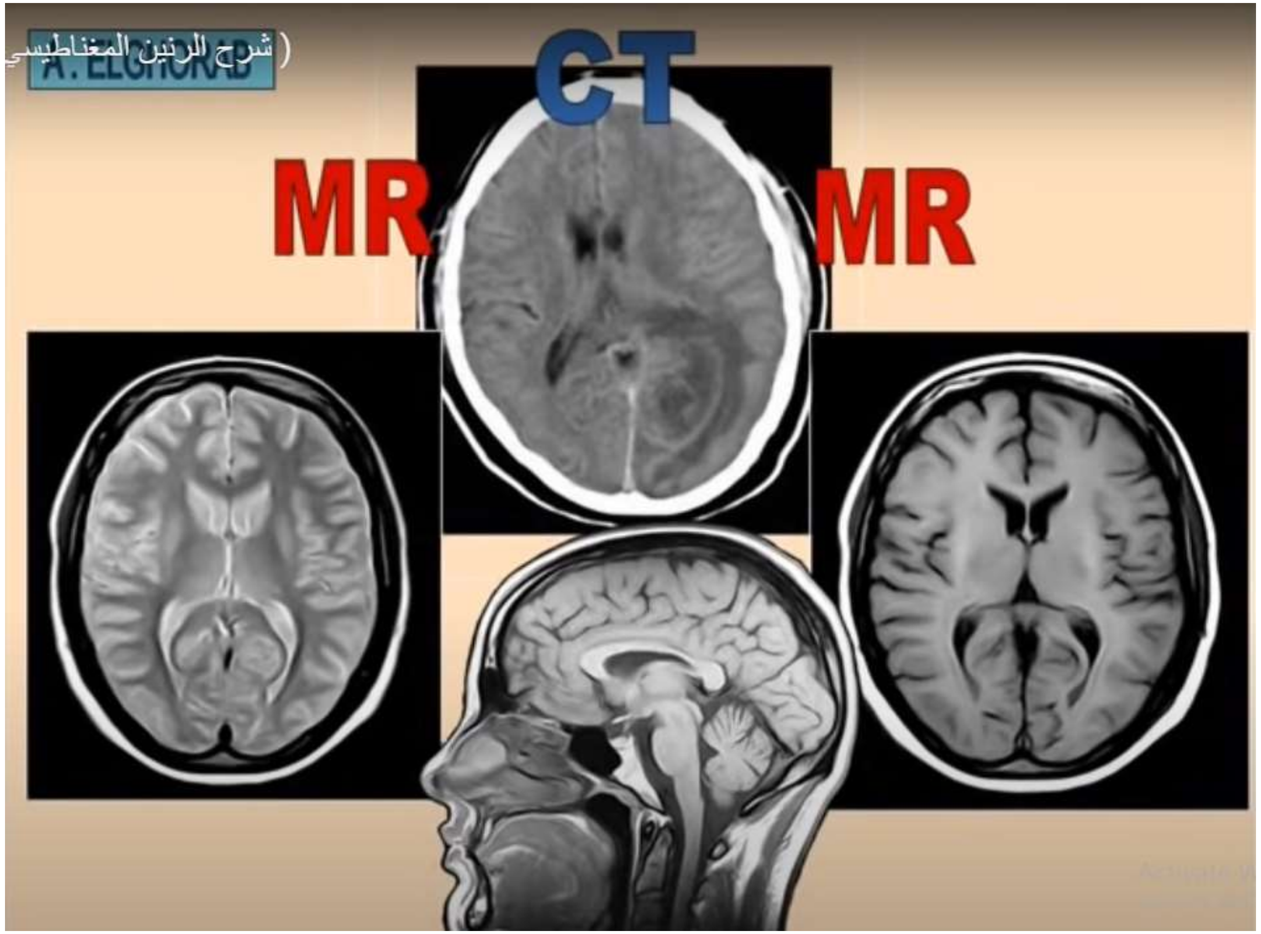
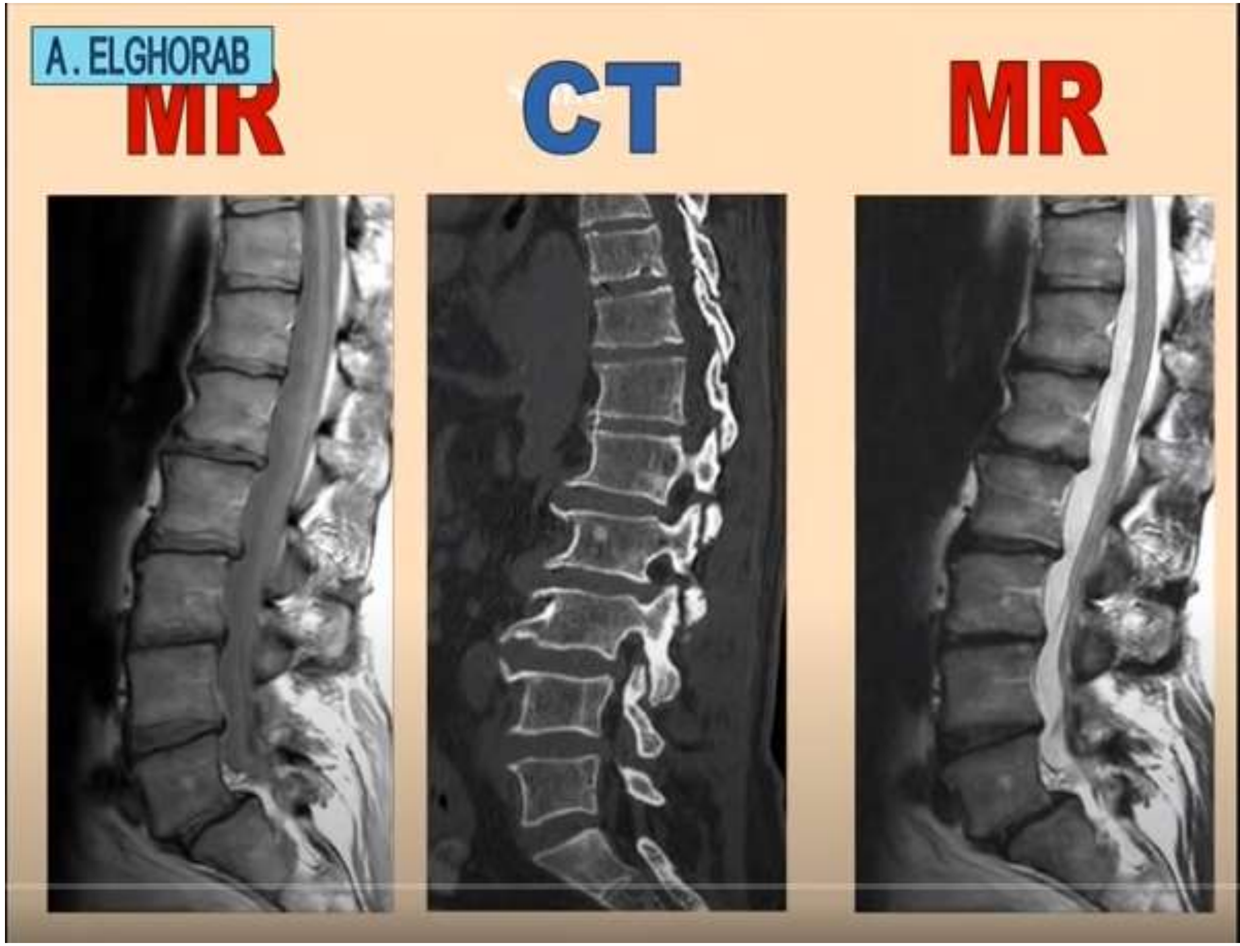
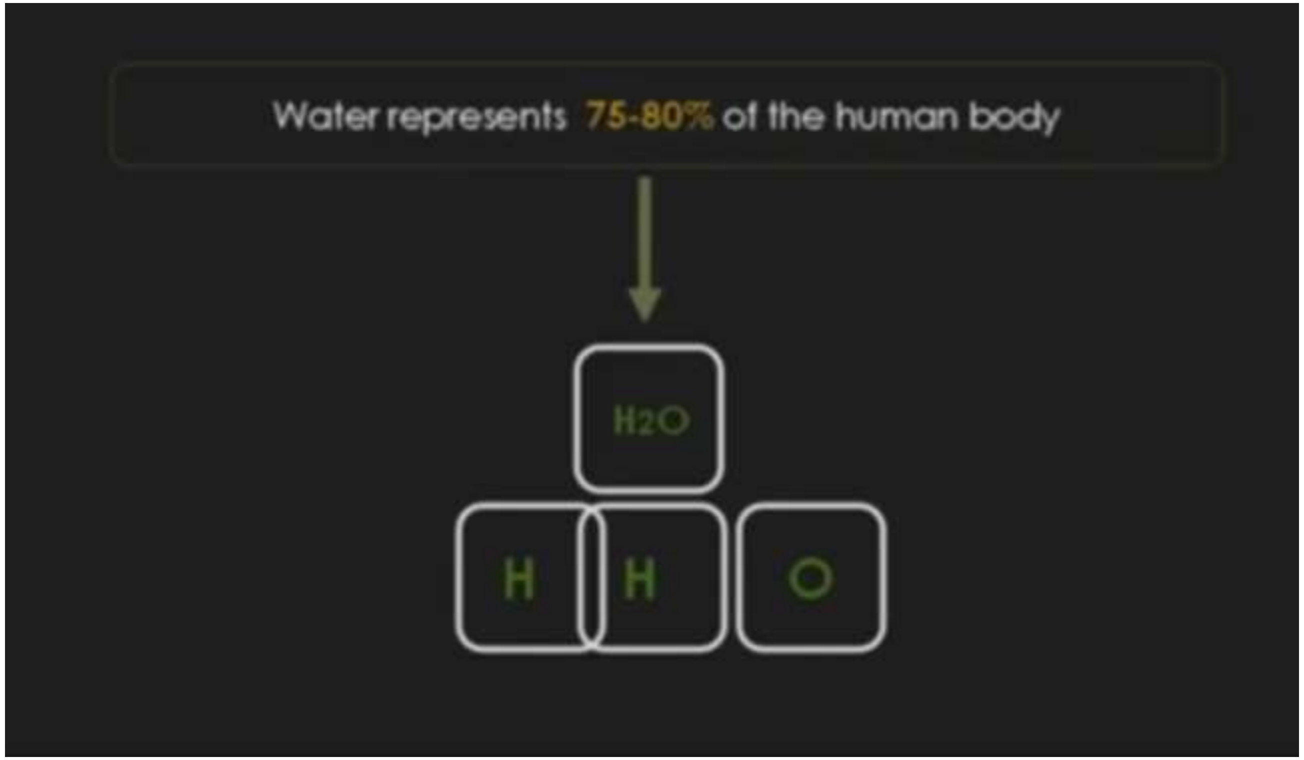
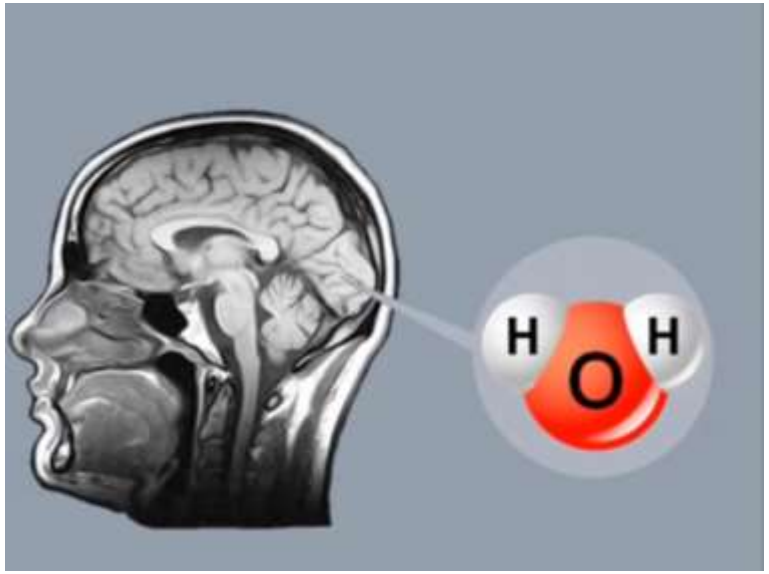
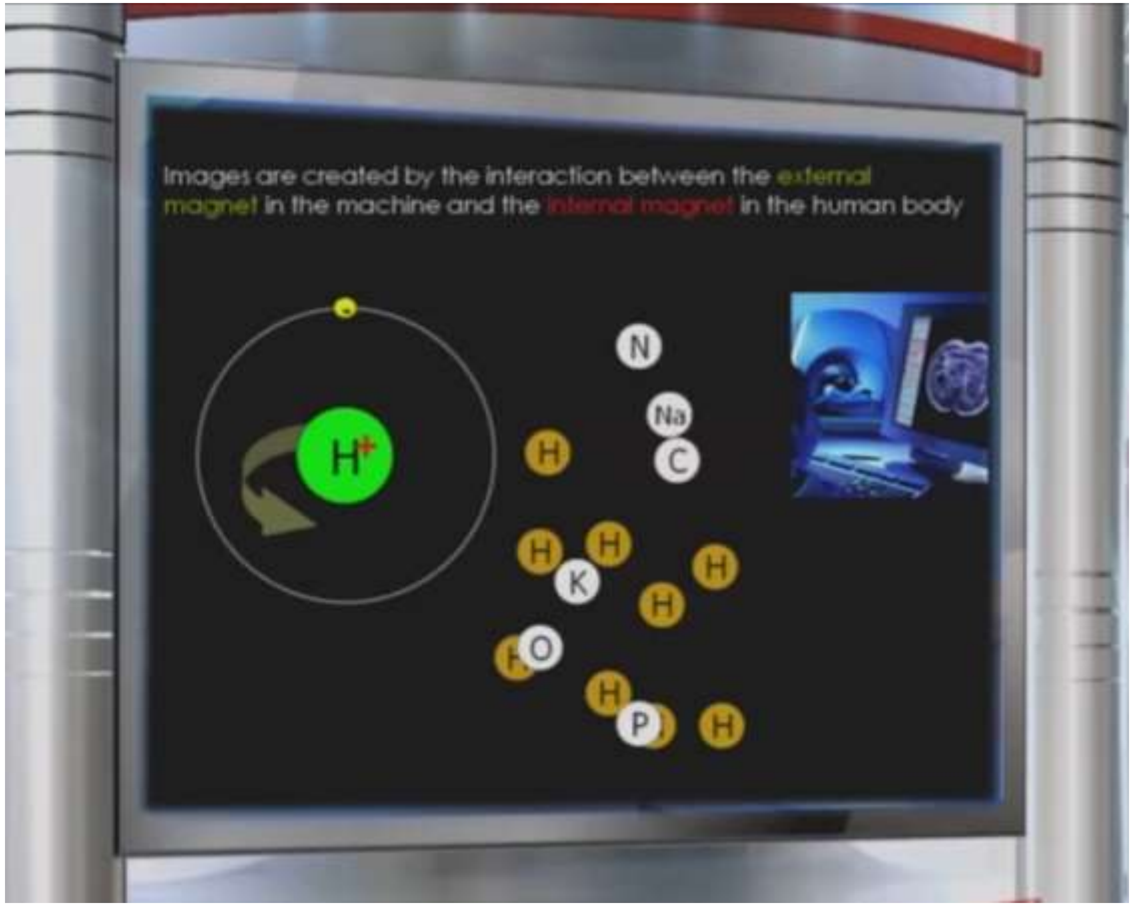

شرح الرنين المغناطيسي )
CT
MR
MR

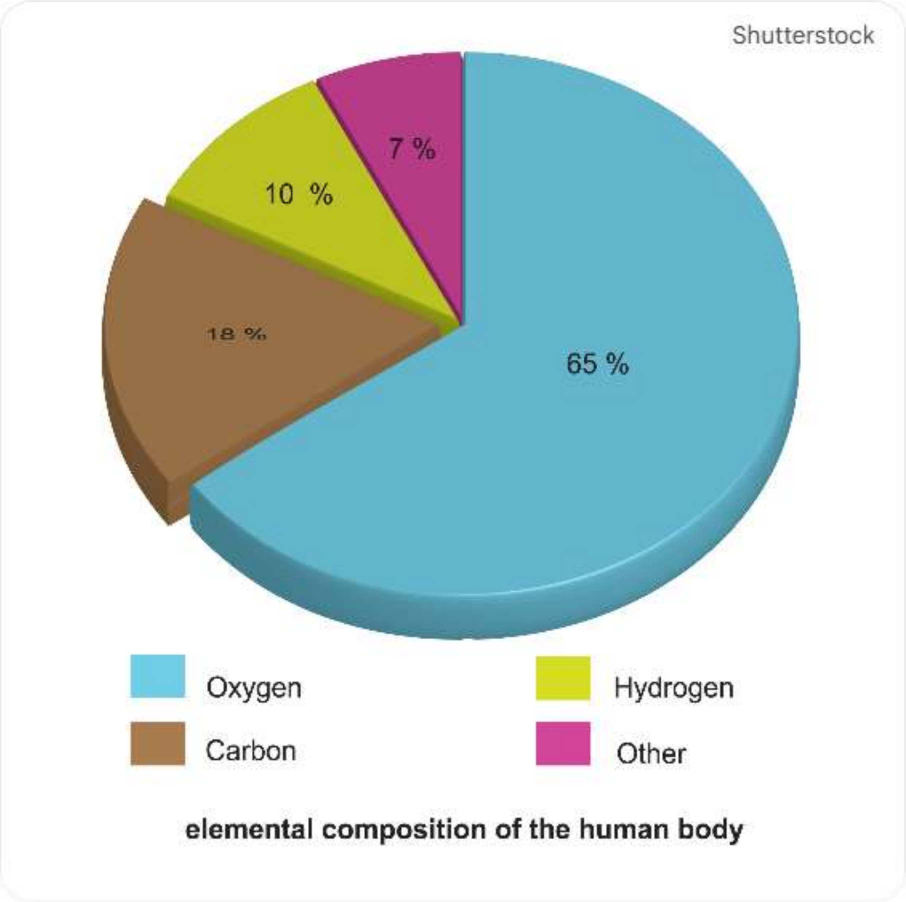

A. ELGHORAB
MR
CT
MR


Water represents 75-80% of the human body
H2O
H
H
O

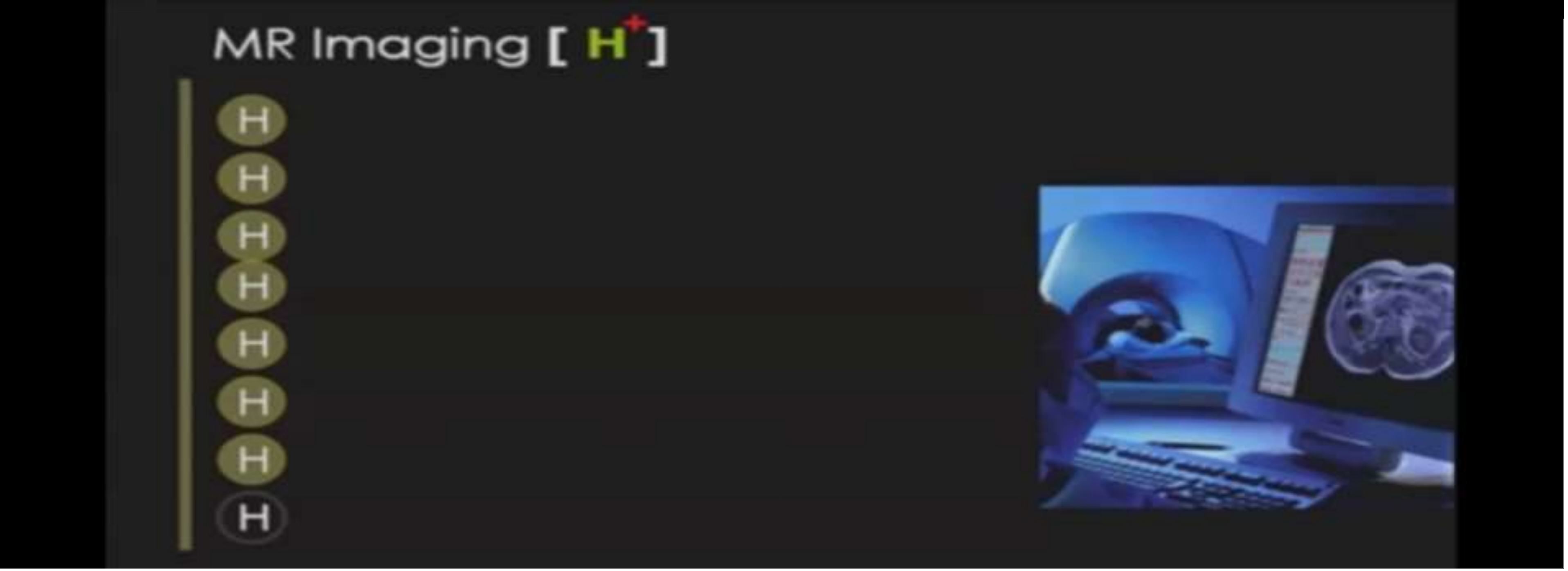

H
O
H

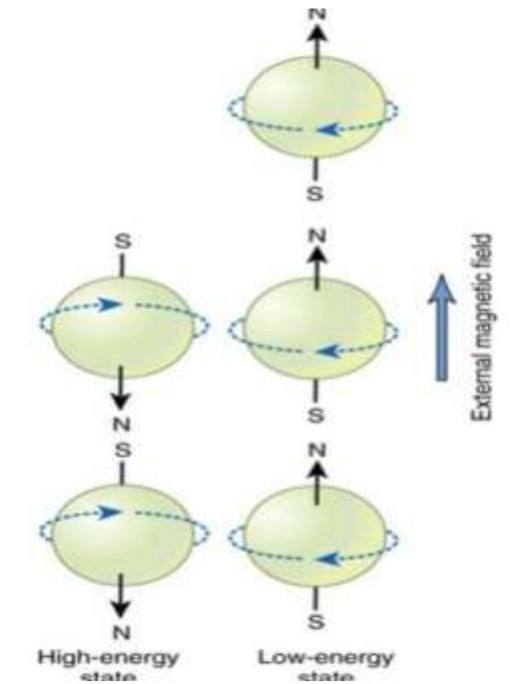


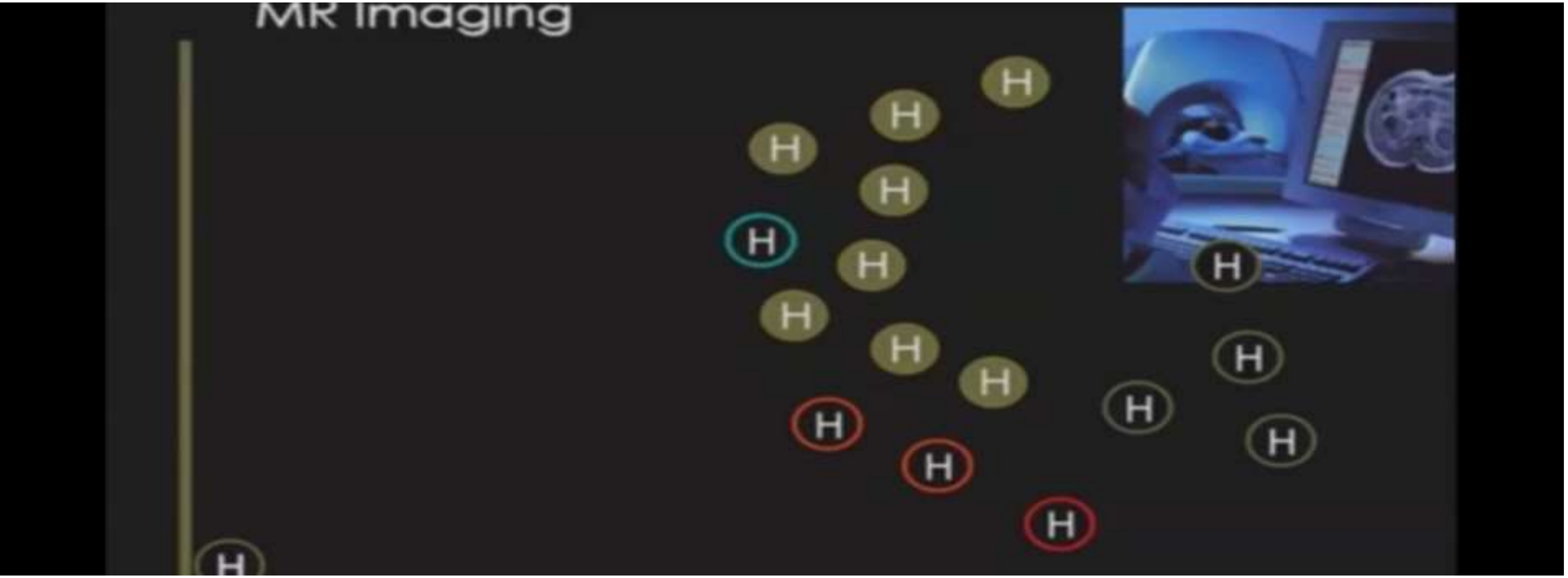
Images are created by the interaction between the external magnet in the machine and the internal magnet in the human body
H+
N
Na
C
H
K
O
P

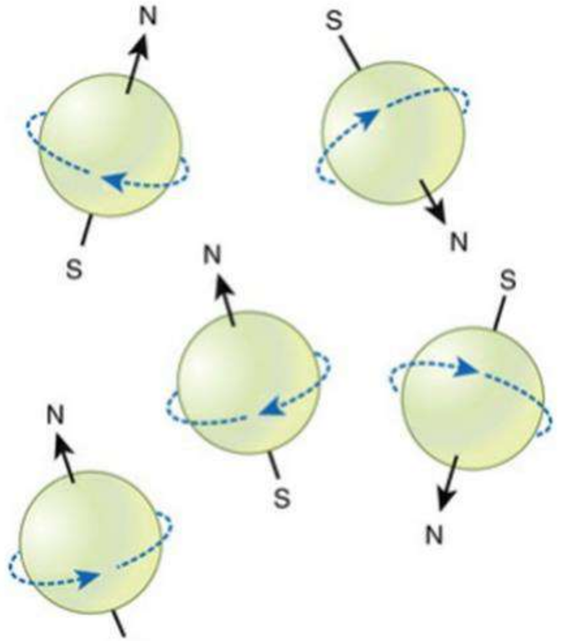



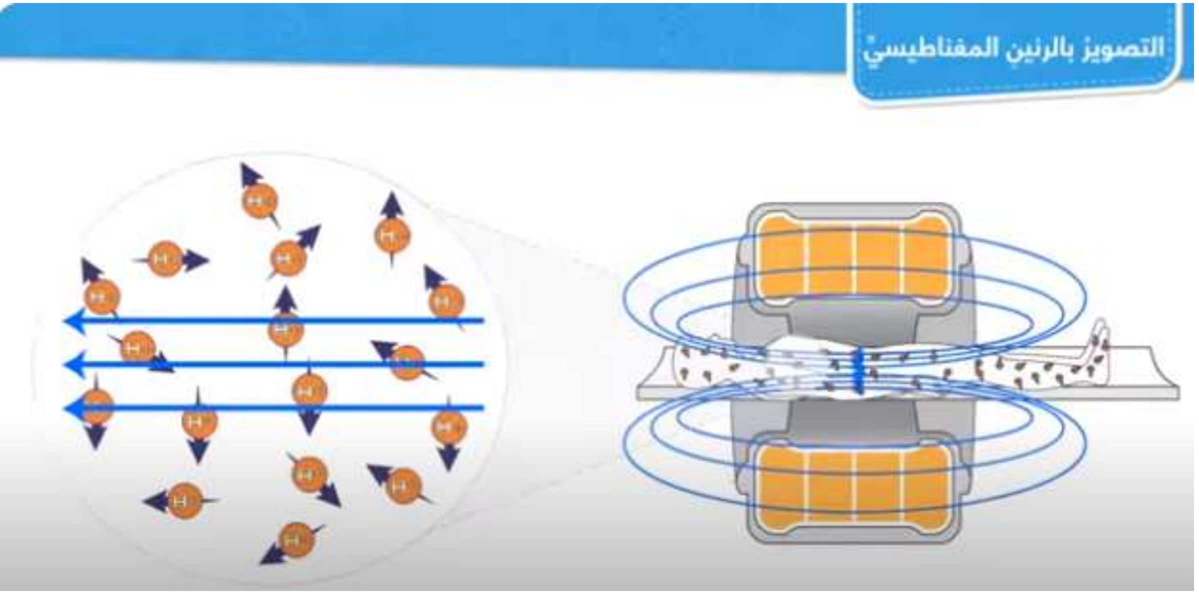
Shutterstock
7 %
10 %
18 %
65 %
Oxygen
Hydrogen
Carbon
Other
elemental composition of the human body

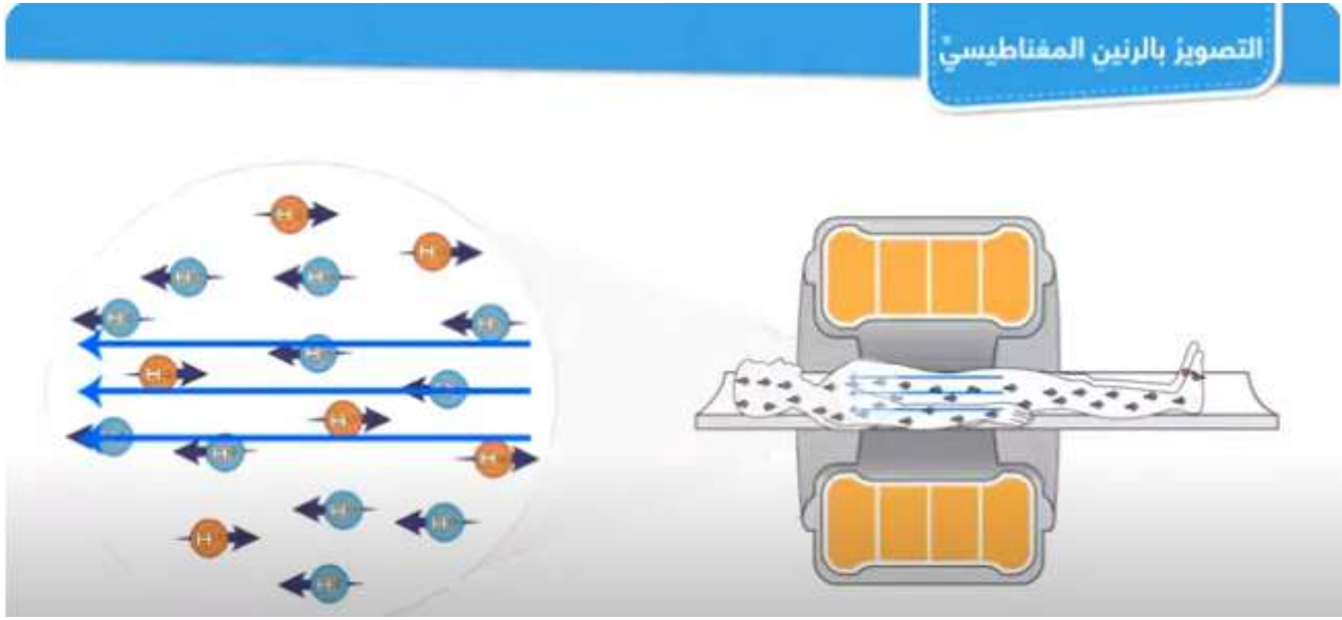

MR Imaging [ H+ ]
H
H
H
H
H
H
H
H


N
S
External magnetic field
High-energy state
Low-energy state


MR Imaging


N
S


التصوير بالرنين المغناطيسي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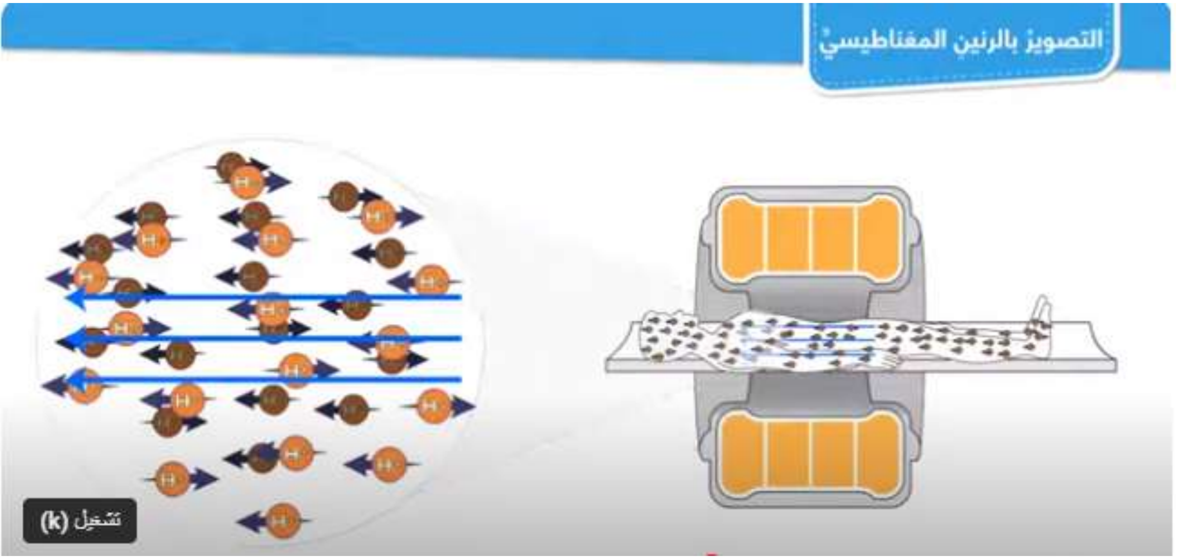
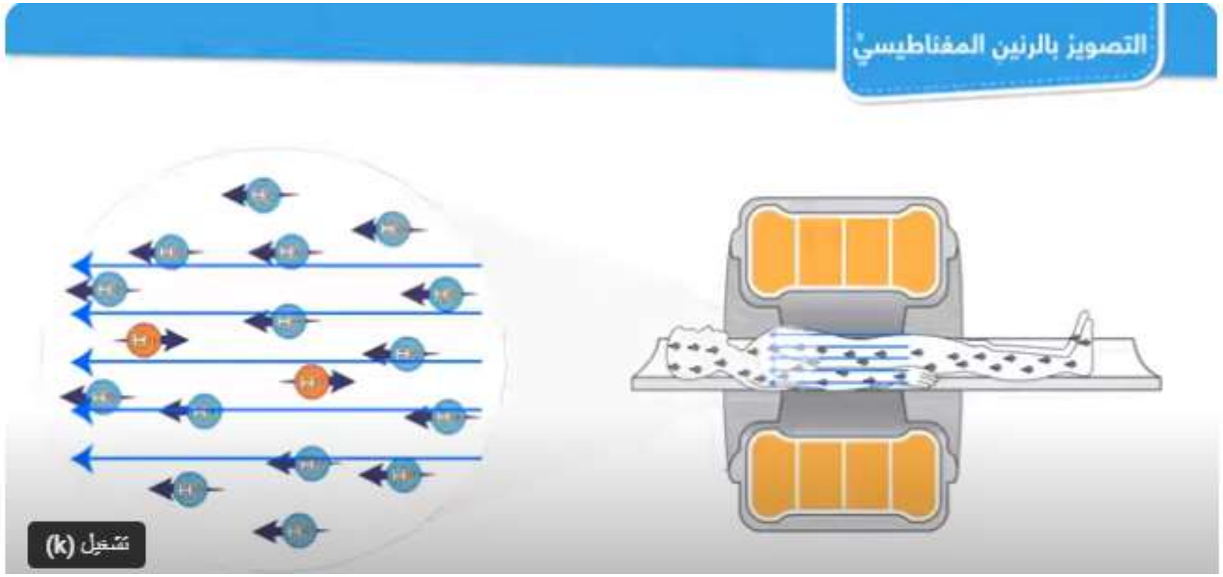

التصوير بالرنين المغناطيسي
تشغيل (k)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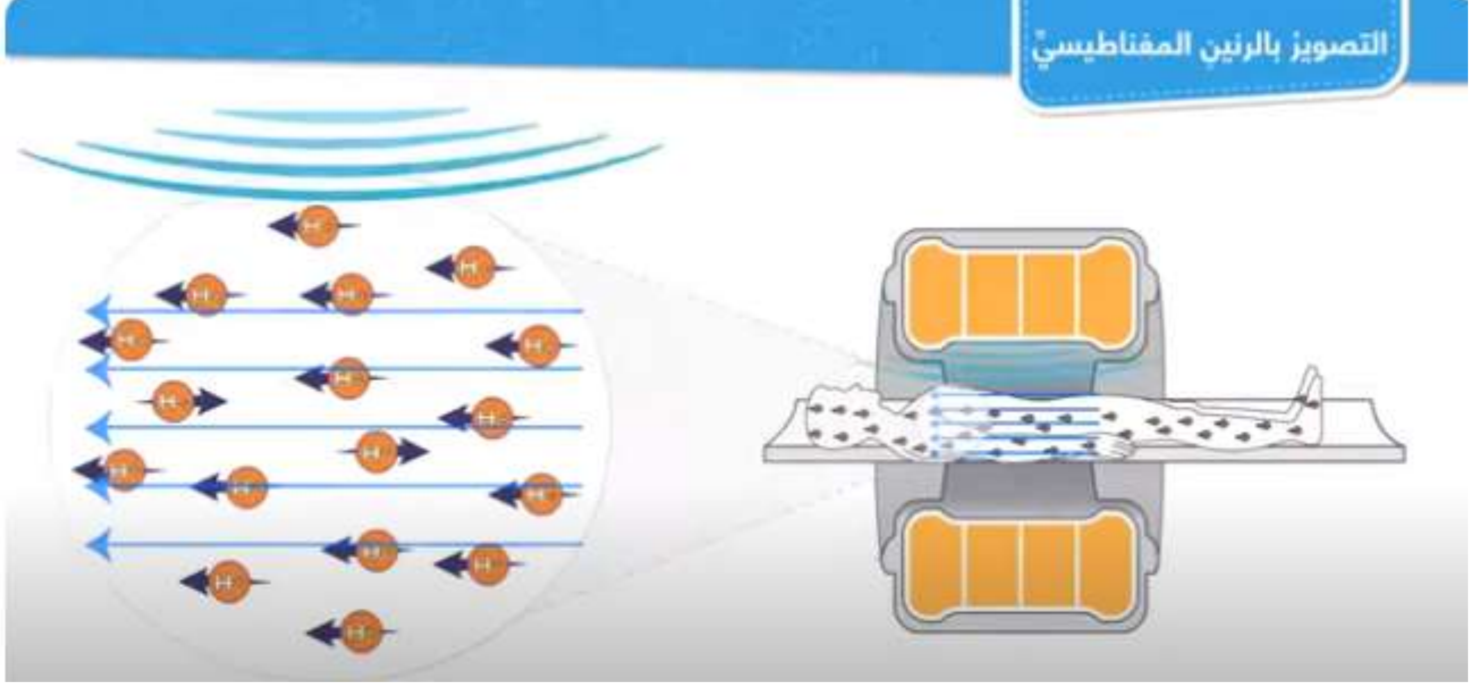
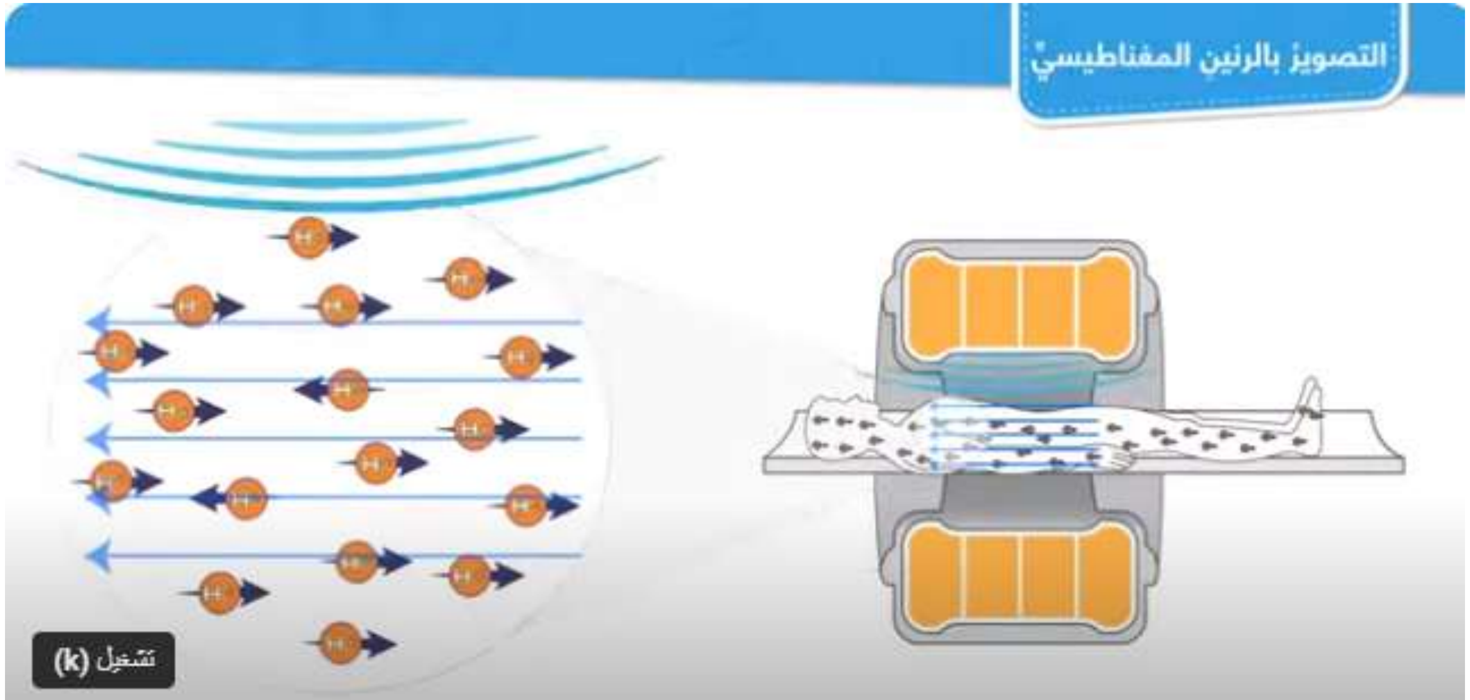
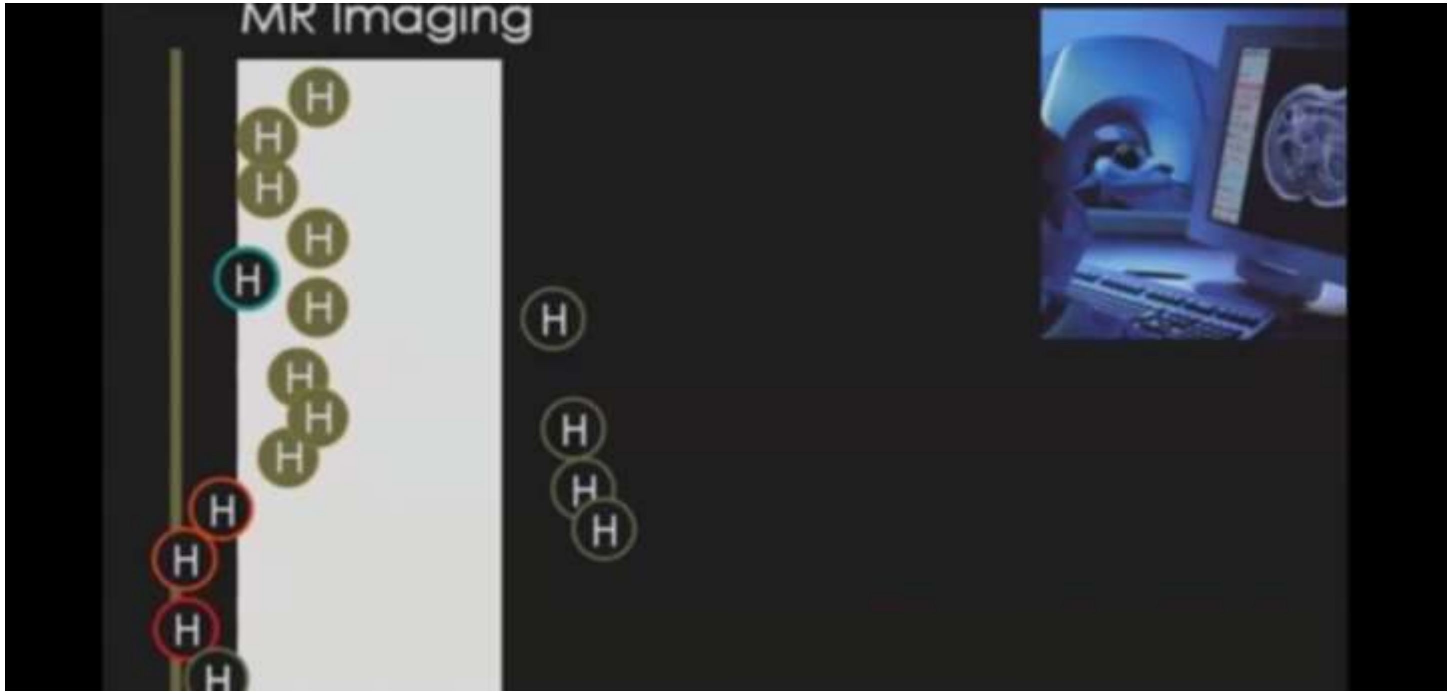
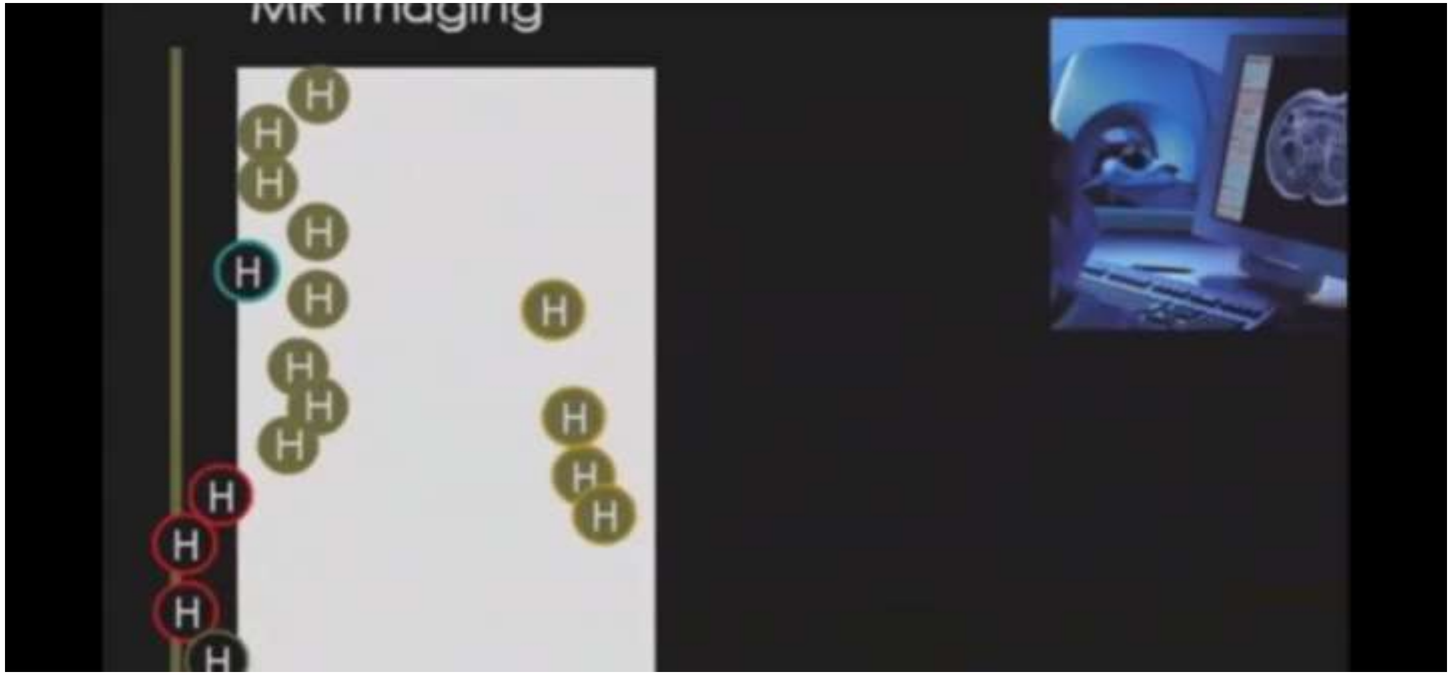
تشغيل (k)


التصويرُ بالرنين المغناطيسيّ

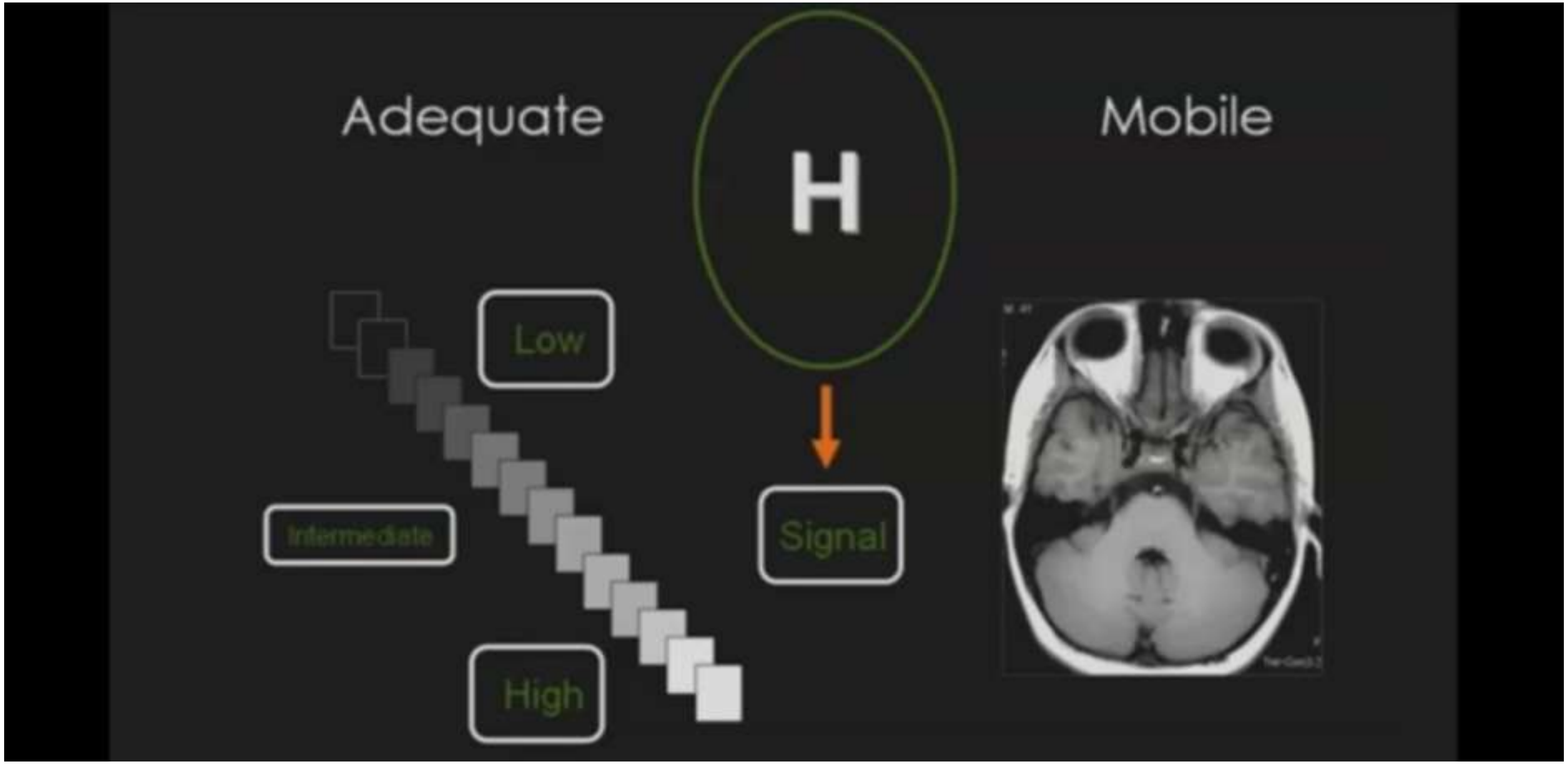

التصويرُ بالرنين المغناطيسيّ
تشغيل (k)



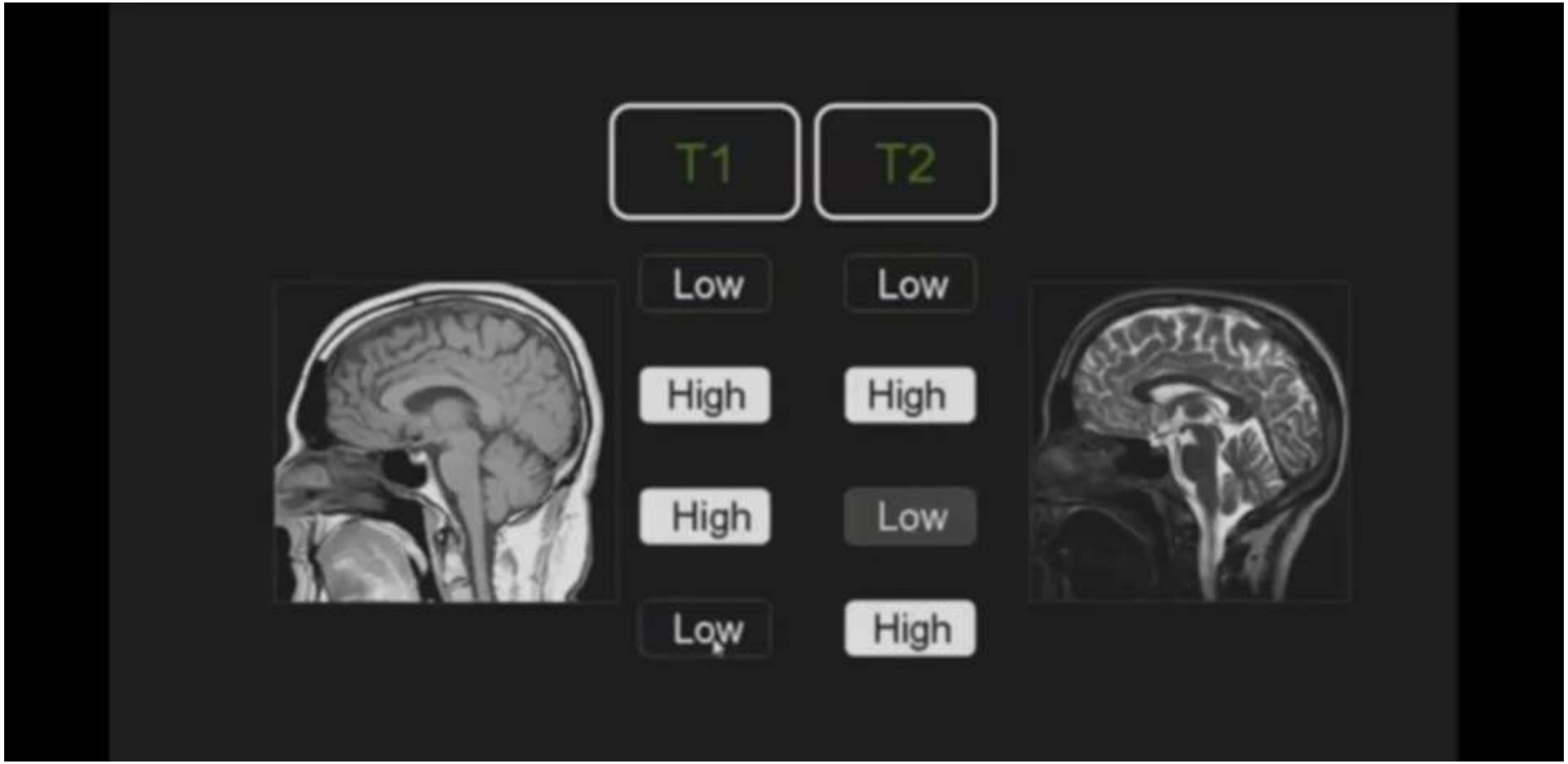

T1




T2



Adequate
H
Mobile
Low
Intermediate
High
Signal

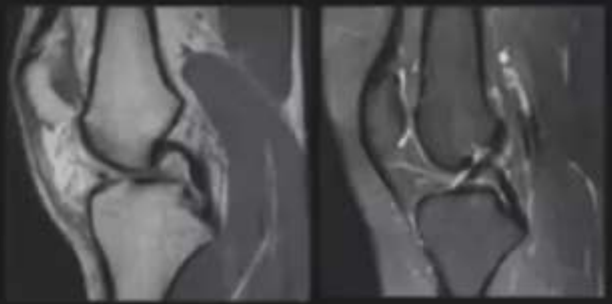


T1
T2
Low
Low
High
High
High
Low
Low
High

T1 LOW | T2 LOW

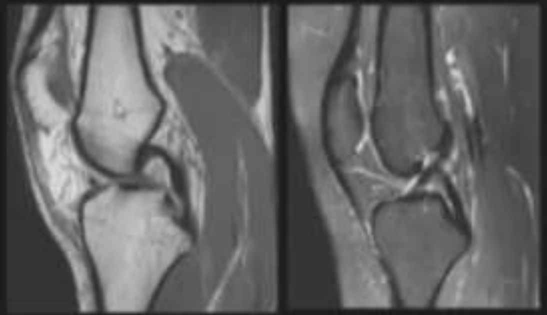
[ Non mobile protons ]

- Cortical bone
- Mature fibrous tissue ( ligaments and tendons)
- Calcifications ( physiological, pathological)



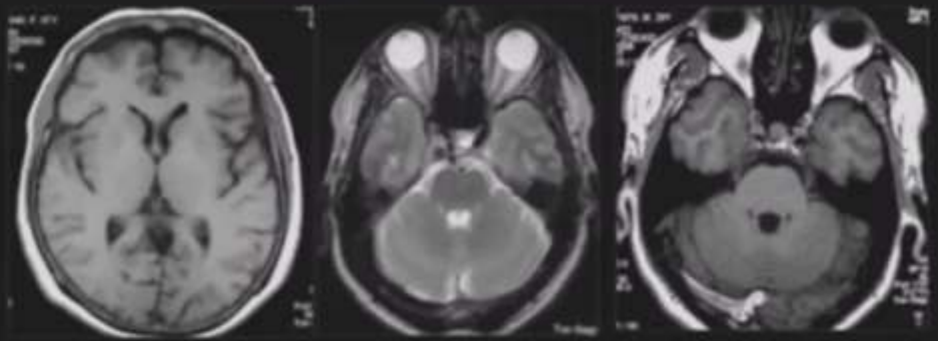
T1 High | T2 Low

Fat [ Subcutaneous fat, bone marrow , dermoid cyst,...]



T1 LOW | T2 High

Fluid [ CSF , effusion , ascites , urine , vitreous ,.. , .... ]

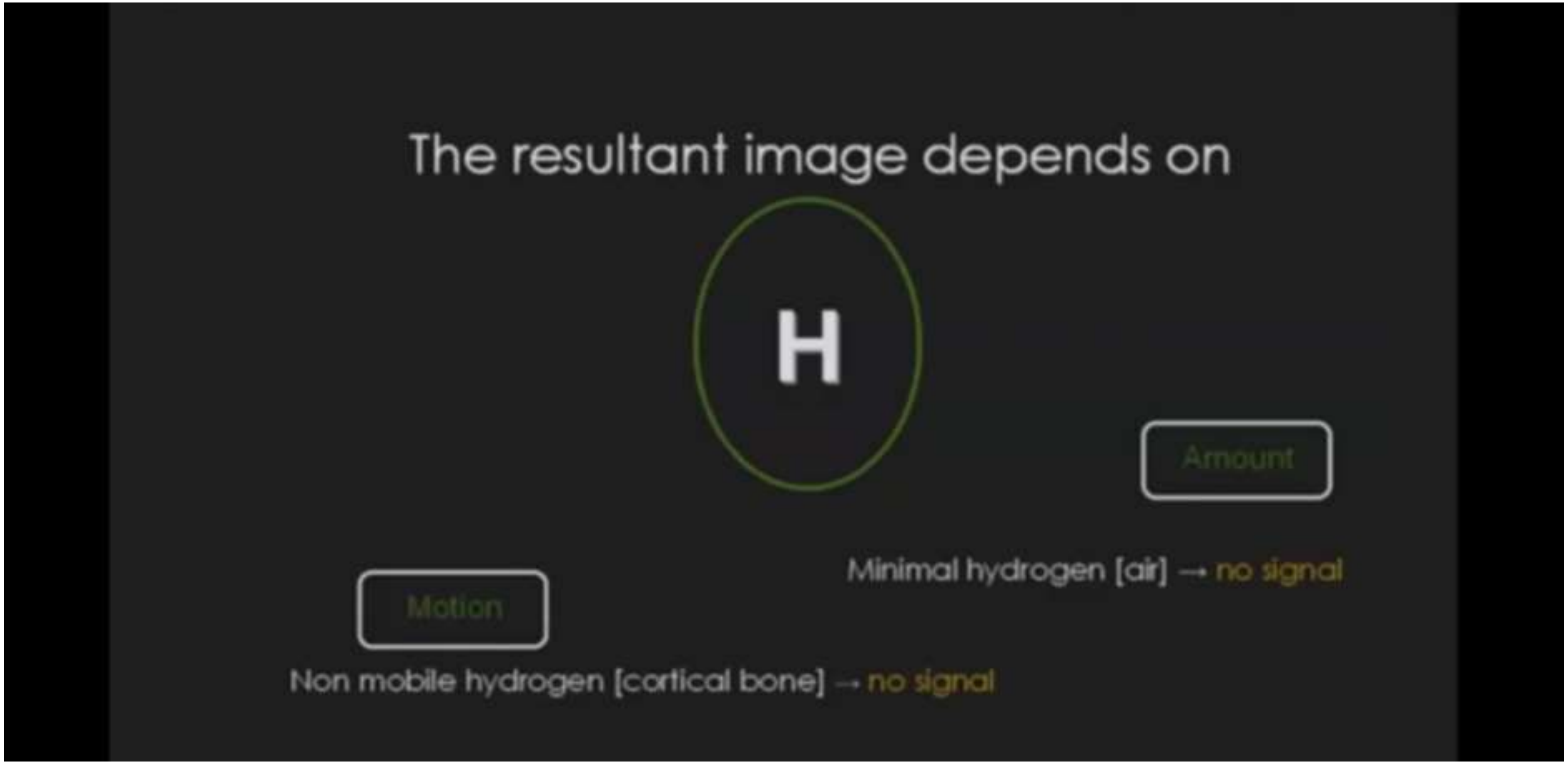


T1 High | T2 High

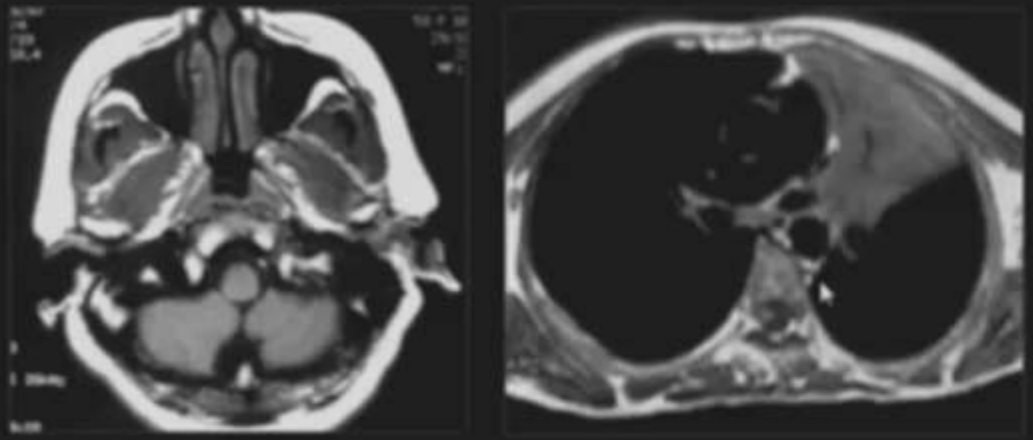
Blood [ Subacute hemorrhage,...]

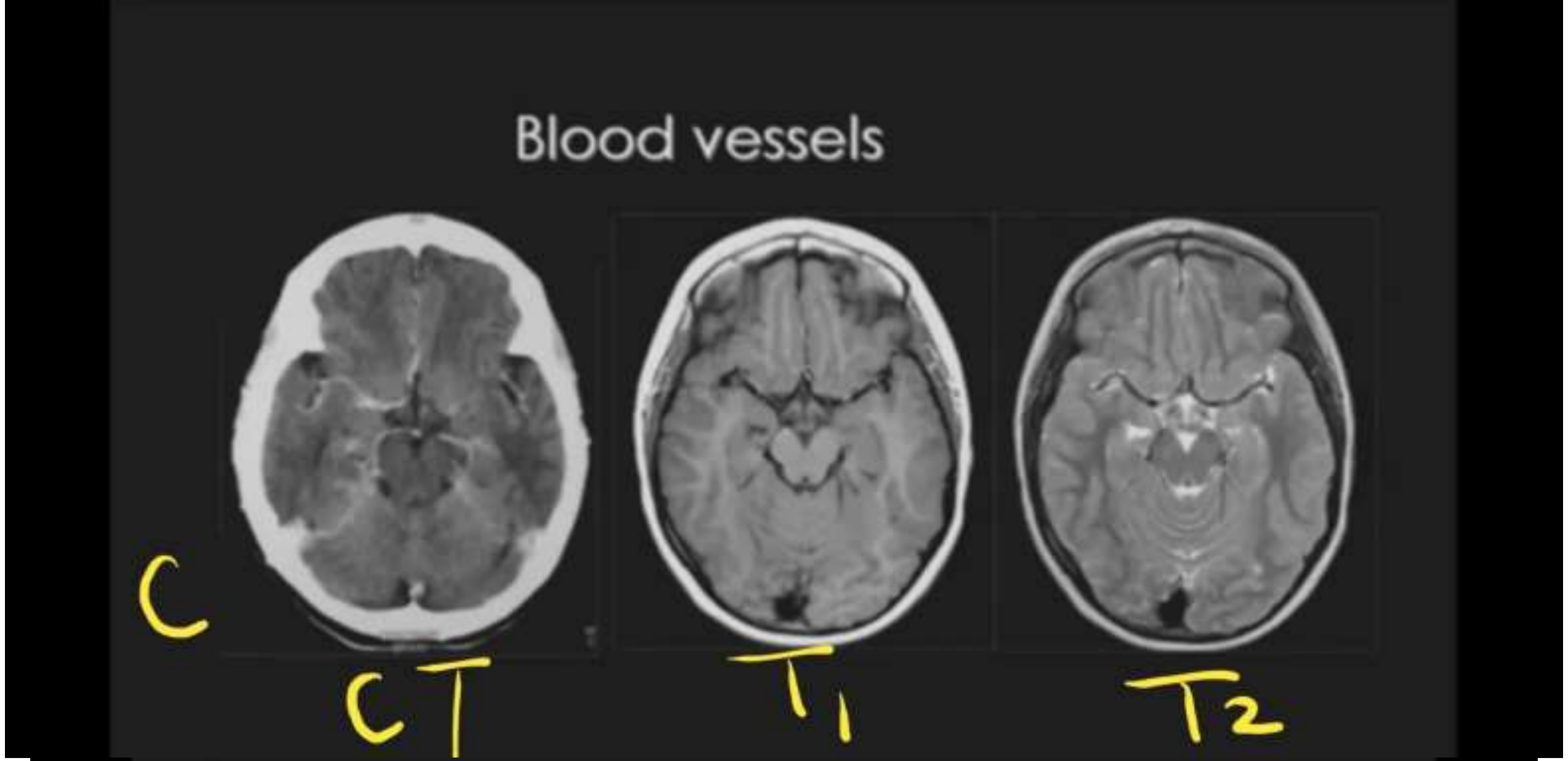
Sub acute epidural hematoma

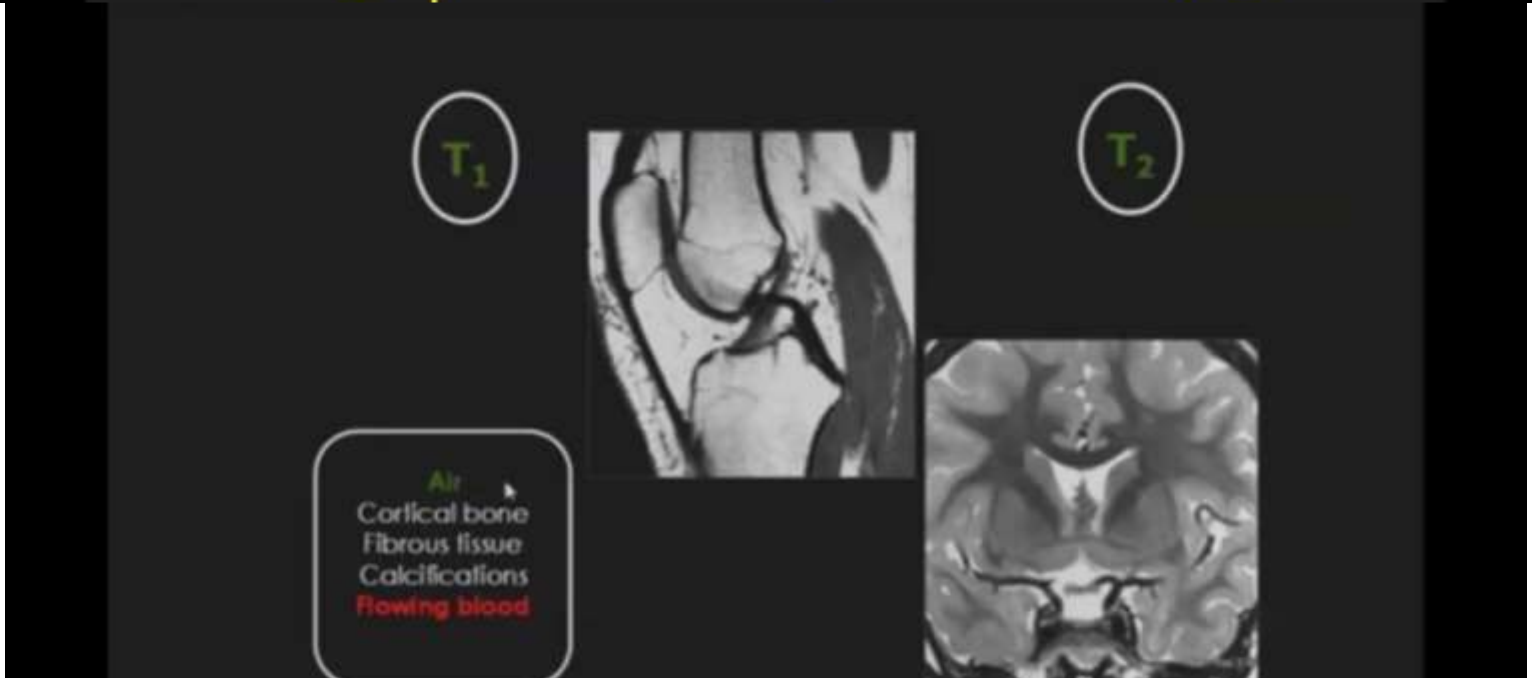
R:740
E:16/Ef
.0thk/1.5sp
:2000
:96/Ef
.0thk/1.5sp

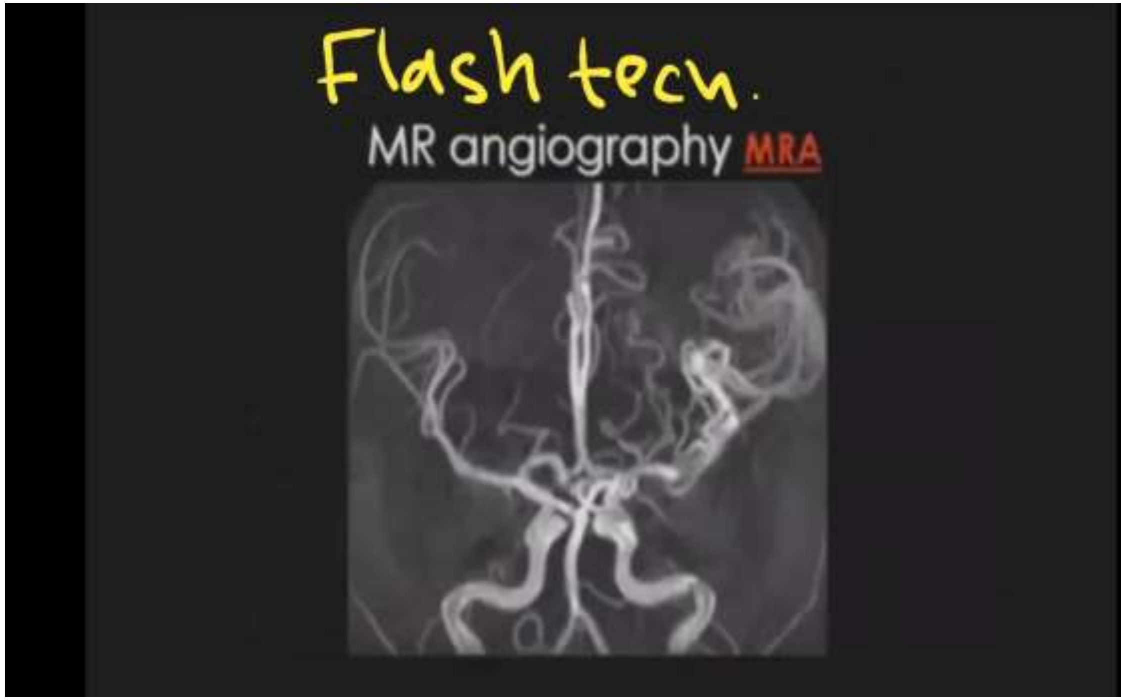

The resultant image depends on
H
Amount
Minimal hydrogen [air] → no signal
Motion
Non mobile hydrogen [cortical bone] → no signal

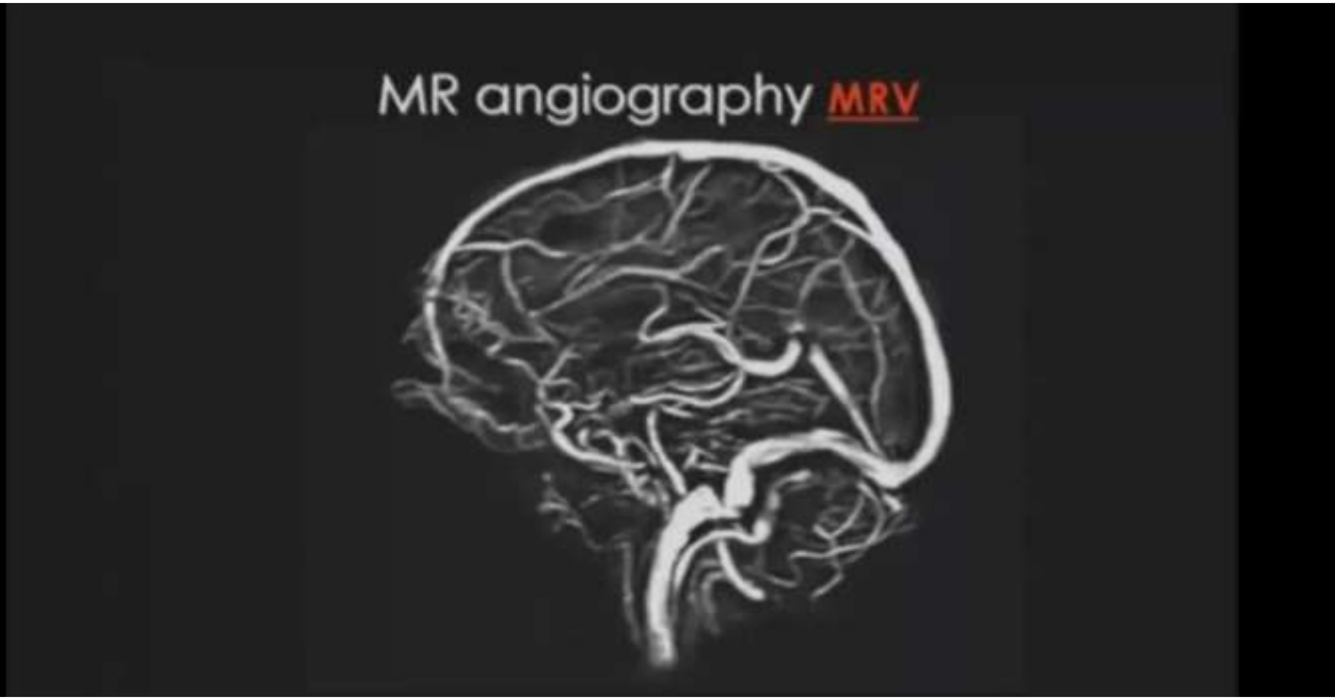
# Minimal hydrogen [air]



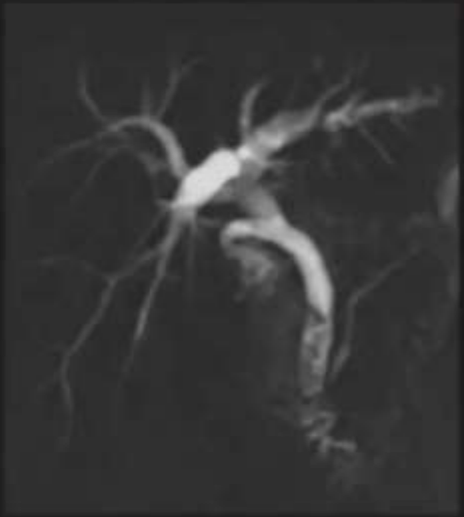

Blood vessels
C
CT
T1
T2


T1
T2
Air
Cortical bone
Fibrous tissue
Calcifications
Flowing blood


Flash tech.
MR angiography MRA


MR angiography MRV

# MR pancreatico-cholangiography



MRCP

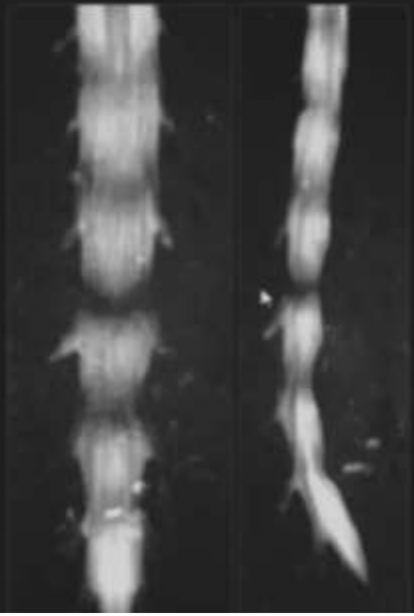
MR urography MRU

No contrast is needed

# MR pancreatico-cholangiography

MRCP

MR Myelography
MRM



- **Advantages:**
- It provides the best contrast resolution for soft tissues.
- It does not involve any harmful ionizing radiation.
- It allows for direct multi-planar imaging without needing to reposition the patient.
- **Disadvantages & Safety Concerns:**
- Imaging times can be long, sometimes reaching up to 30 minutes.
- The procedure carries a high imaging cost.
- The narrow space and acoustic noise can cause claustrophobia or discomfort. This can be managed using open MRIs, sedation, anesthesia, or headphones playing music.
- Ferromagnetic metals inside the body (e.g., pacemakers, insulin pumps, cerebral aneurysm clips, shrapnel) are major hazards. The magnetic field can move these objects, cause excessive heating, or induce strong electrical currents. Note: Gold, stainless steel, and iron oxide are ferromagnetic, while titanium and amalgam are not.
- Tattoos containing iron oxide (found in some red, black, brown, or flesh-colored dyes) can cause a burning sensation because the magnetic metals can convert RF pulses into electricity or be "pulled" by the magnet.
- Gadolinium contrast agents should be avoided during pregnancy because they cross the placenta and enter the fetal bloodstream.

- **Clinical Applications**
- Due to its excellent soft tissue resolution, MRI is highly effective for evaluating:
- The integrity and position of the disk in the temporomandibular joint (TMJ).
- Soft tissue diseases, particularly neoplasia of the neck, cheek, tongue, and salivary glands.
- Malignant involvement in lymph nodes.
- Suspected early osteomyelitis of the jaws to detect edematous changes in fatty marrow and surrounding soft tissue.
- Blood flow in arteries (via MR angiography) to diagnose aneurysms, occlusions, or malformations

# US UltraSound
# B-mode britghness mode, Doppler color mode

- Anechoic
- Hypoechoic
- Hyperechoic
- diagnostic
- Periodontology, implantology
- Oral and MFS
- Oral pathology
- Salivary gland
- endodontics
- Therapeutic Applications
- Ultrasonic Scaling (calculus)
- Tissue Healing & Pain Management
- "micro-massage."
- Endodontic Cleaning